

```
In [260... import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns
sns.set()

pd.set_option('display.max_rows', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.width', None)
pd.set_option('display.max_colwidth', None)
```

Load In and Explore Dataset

```
In [288... #read in dataset file and create dataframe

file = "C:\\Users\\Ivan\\Documents\\MSDS422\\Assignment 1\\HMEQ_Loss.csv"

df = pd.read_csv( file )
```

```
In [289... #Display first five rows with rows and columns TRANSPOSED

print(df.head(5).T)
```

	0	1	2	3	4
TARGET_BAD_FLAG	1	1	1	1	0
TARGET_LOSS_AMT	641.0	1109.0	767.0	1425.0	NaN
LOAN	1100	1300	1500	1500	1700
MORTDUE	25860.0	70053.0	13500.0	NaN	97800.0
VALUE	39025.0	68400.0	16700.0	NaN	112000.0
REASON	HomeImp	HomeImp	HomeImp	NaN	HomeImp
JOB	Other	Other	Other	NaN	Office
YOJ	10.5	7.0	4.0	NaN	3.0
DEROG	0.0	0.0	0.0	NaN	0.0
DELINQ	0.0	2.0	0.0	NaN	0.0
CLAGE	94.366667	121.833333	149.466667	NaN	93.333333
NINQ	1.0	0.0	1.0	NaN	0.0
CLNO	9.0	14.0	10.0	NaN	14.0
DEBTINC	NaN	NaN	NaN	NaN	NaN

```
In [290... #explore the field types

datatypes = df.dtypes
print(datatypes)
```

```

TARGET_BAD_FLAG      int64
TARGET_LOSS_AMT      float64
LOAN                  int64
MORTDUE              float64
VALUE                float64
REASON               object
JOB                  object
YOJ                  float64
DEROG                float64
DELINQ               float64
CLAGE                float64
NINQ                 float64
CLNO                 float64
DEBTINC              float64
dtype: object

```

```

In [291... #display the description statistics of the data (transposed)

descript = df.describe(include="all").T
print(descript)

```

	count	unique	top	freq	mean	std \
TARGET_BAD_FLAG	5960.0	NaN	NaN	NaN	0.199497	0.399656
TARGET_LOSS_AMT	1189.0	NaN	NaN	NaN	13414.576955	10839.455965
LOAN	5960.0	NaN	NaN	NaN	18607.969799	11207.480417
MORTDUE	5442.0	NaN	NaN	NaN	73760.8172	44457.609458
VALUE	5848.0	NaN	NaN	NaN	101776.048741	57385.775334
REASON	5708	2	DebtCon	3928	NaN	NaN
JOB	5681	6	Other	2388	NaN	NaN
YOJ	5445.0	NaN	NaN	NaN	8.922268	7.573982
DEROG	5252.0	NaN	NaN	NaN	0.25457	0.846047
DELINQ	5380.0	NaN	NaN	NaN	0.449442	1.127266
CLAGE	5652.0	NaN	NaN	NaN	179.766275	85.810092
NINQ	5450.0	NaN	NaN	NaN	1.186055	1.728675
CLNO	5738.0	NaN	NaN	NaN	21.296096	10.138933
DEBTINC	4693.0	NaN	NaN	NaN	33.779915	8.601746

	min	25%	50%	75%	max
TARGET_BAD_FLAG	0.0	0.0	0.0	0.0	1.0
TARGET_LOSS_AMT	224.0	5639.0	11003.0	17634.0	78987.0
LOAN	1100.0	11100.0	16300.0	23300.0	89900.0
MORTDUE	2063.0	46276.0	65019.0	91488.0	399550.0
VALUE	8000.0	66075.5	89235.5	119824.25	855909.0
REASON	NaN	NaN	NaN	NaN	NaN
JOB	NaN	NaN	NaN	NaN	NaN
YOJ	0.0	3.0	7.0	13.0	41.0
DEROG	0.0	0.0	0.0	0.0	10.0
DELINQ	0.0	0.0	0.0	0.0	15.0
CLAGE	0.0	115.116702	173.466667	231.562278	1168.233561
NINQ	0.0	0.0	1.0	2.0	17.0
CLNO	0.0	15.0	20.0	26.0	71.0
DEBTINC	0.524499	29.140031	34.818262	39.003141	203.312149

Organize Fields and Explore by Field Type

In [292...

```
#organize fields into lists by type
#leaving out the two target variables

dt = df.dtypes

TARGET_F = "TARGET_BAD_FLAG"
TARGET_A = "TARGET_LOSS_AMT"

objList = []
intList = []
floatList = []

for i in dt.index:
    if i in ([TARGET_F, TARGET_A]):continue
    if dt[i] in (["object"]):objList.append(i)
    if dt[i] in (["float64"]):floatList.append(i)
    if dt[i] in (["int64"]):intList.append(i)

for i in objList:
    print(i)
```

REASON

JOB

In [293...

```
for i in floatList:
    print(i)
```

MORTDUE

VALUE

YOJ

DEROG

DELINQ

CLAGE

NINQ

CLNO

DEBTINC

In [294...

```
for i in intList:
    print(i)
```

LOAN

In [295...

```
#explore the categorical(object) variables

for i in objList:
    print(" Field = ", i)
    g = df.groupby(i)
    print(g[i].count())
    x = g[TARGET_F].mean()
    print("Unpaid Loan Probability: ", x)
    y = g[TARGET_A].mean()
    print("Avg Unpaid Amt: ", y)
```

```

Field = REASON
REASON
DebtCon    3928
HomeImp    1780
Name: REASON, dtype: int64
Unpaid Loan Probability: REASON
DebtCon    0.189664
HomeImp    0.222472
Name: TARGET_BAD_FLAG, dtype: float64
Avg Unpaid Amt: REASON
DebtCon    16005.163758
HomeImp    8388.090909
Name: TARGET_LOSS_AMT, dtype: float64
Field = JOB
JOB
Mgr         767
Office      948
Other      2388
ProfExe    1276
Sales       109
Self        193
Name: JOB, dtype: int64
Unpaid Loan Probability: JOB
Mgr         0.233377
Office      0.131857
Other       0.231993
ProfExe     0.166144
Sales       0.348624
Self        0.300518
Name: TARGET_BAD_FLAG, dtype: float64
Avg Unpaid Amt: JOB
Mgr        14141.536313
Office     13475.304000
Other     11570.102888
ProfExe    14660.966981
Sales     16421.447368
Self       22232.362069
Name: TARGET_LOSS_AMT, dtype: float64

```

In [296...

#explore the integer variables

```

for i in intList:
    print("Field = ", i)
    g = df.groupby(TARGET_F)
    x = g[i].mean()
    print("Average Loan By ", x)
    c = df[i].corr(df[TARGET_A])
    c = round(100*c, 1)
    print("Amount Unpaid Correlation: ", c, "%")

```

```

Field = LOAN
Average Loan By TARGET_BAD_FLAG
0    19028.107315
1    16922.119428
Name: LOAN, dtype: float64
Amount Unpaid Correlation: 83.7 %

```

In [297...

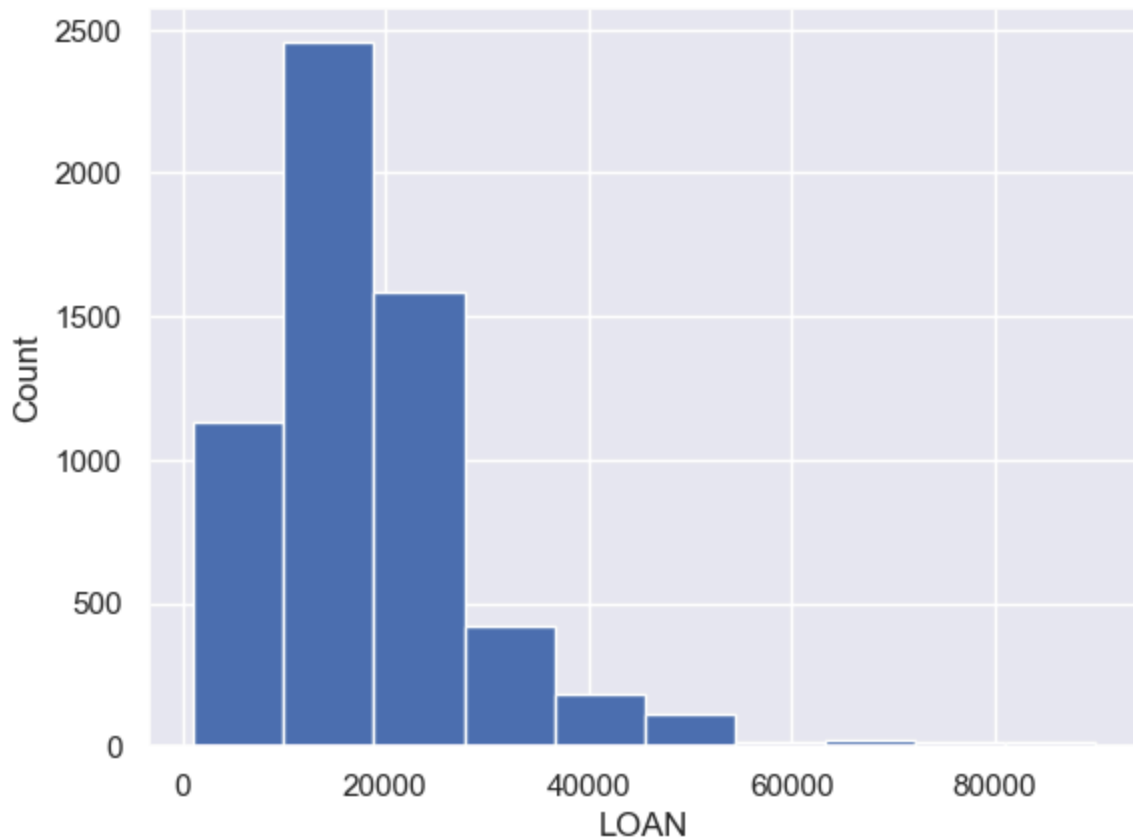
#explore the float variables

```
for i in floatList:
    print("Field = ", i)
    g = df.groupby(TARGET_F)
    x = g[i].mean()
    print("Average Loan By ", x)
    c = df[i].corr(df[TARGET_A])
    c = round(100*c, 1)
    print("Amount Unpaid Correlation: ", c, "%")
```

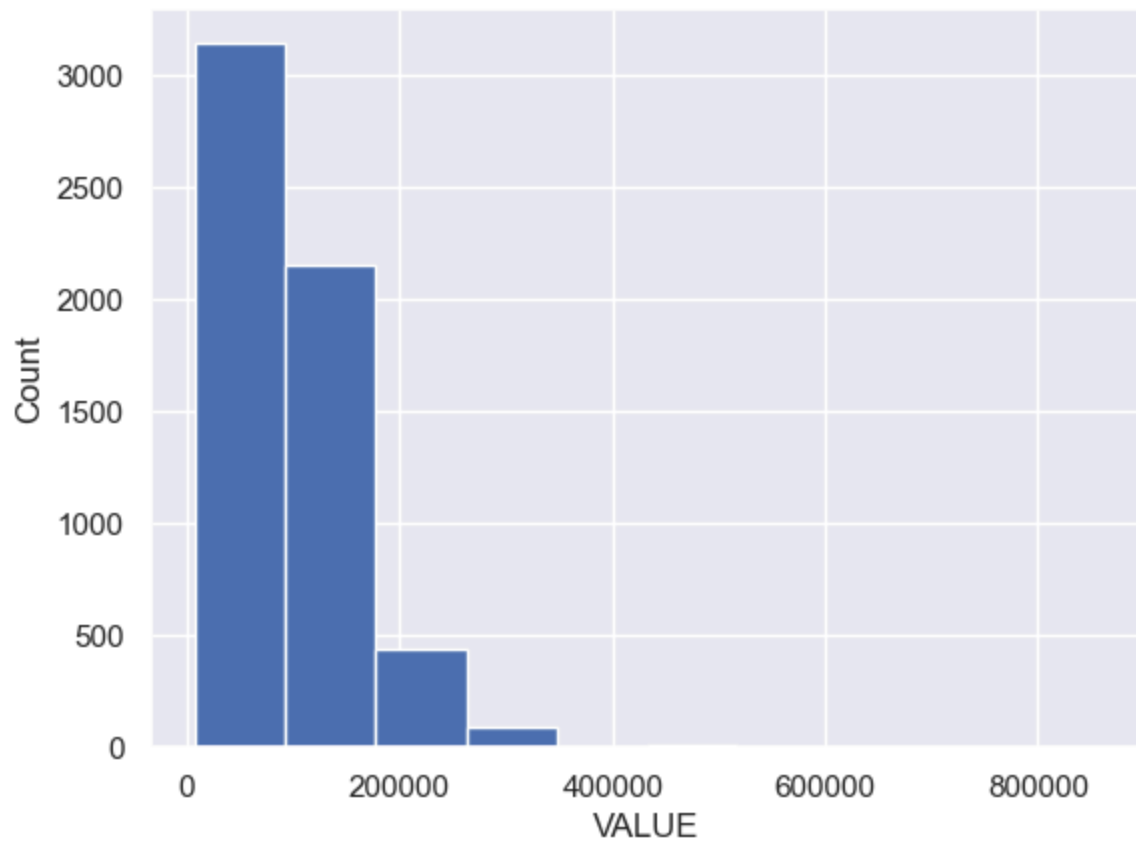
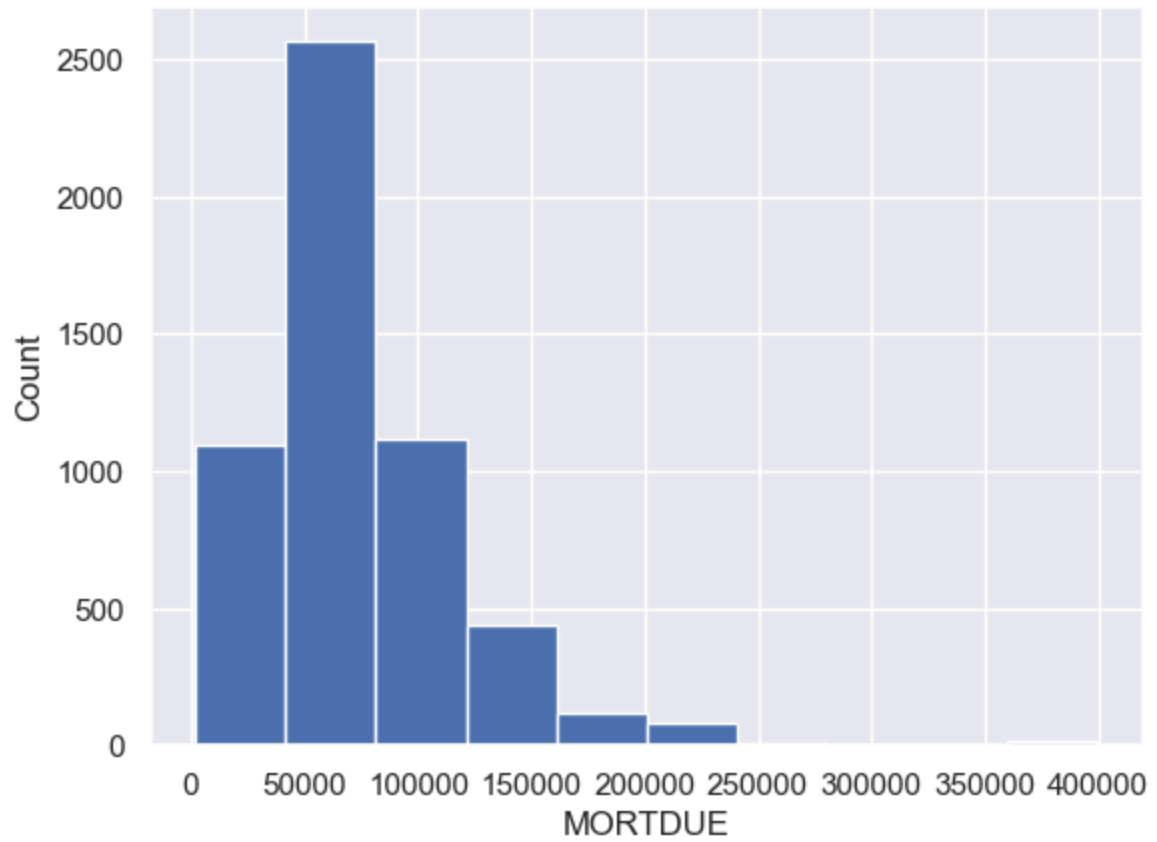
```
Field = MORTDUE
Average Loan By TARGET_BAD_FLAG
0    74829.249055
1    69460.452973
Name: MORTDUE, dtype: float64
Amount Unpaid Correlation: 34.9 %
Field = VALUE
Average Loan By TARGET_BAD_FLAG
0    102595.921018
1     98172.846227
Name: VALUE, dtype: float64
Amount Unpaid Correlation: 36.7 %
Field = YOJ
Average Loan By TARGET_BAD_FLAG
0     9.154941
1     8.027802
Name: YOJ, dtype: float64
Amount Unpaid Correlation: 1.8 %
Field = DEROG
Average Loan By TARGET_BAD_FLAG
0     0.134217
1     0.707804
Name: DEROG, dtype: float64
Amount Unpaid Correlation: 9.5 %
Field = DELINQ
Average Loan By TARGET_BAD_FLAG
0     0.245133
1     1.229185
Name: DELINQ, dtype: float64
Amount Unpaid Correlation: 22.7 %
Field = CLAGE
Average Loan By TARGET_BAD_FLAG
0    187.002355
1    150.190183
Name: CLAGE, dtype: float64
Amount Unpaid Correlation: 1.4 %
Field = NINQ
Average Loan By TARGET_BAD_FLAG
0     1.032749
1     1.782765
Name: NINQ, dtype: float64
Amount Unpaid Correlation: 14.3 %
Field = CLNO
Average Loan By TARGET_BAD_FLAG
0    21.317036
1    21.211268
Name: CLNO, dtype: float64
Amount Unpaid Correlation: 40.0 %
Field = DEBTINC
Average Loan By TARGET_BAD_FLAG
0    33.253129
1    39.387645
Name: DEBTINC, dtype: float64
Amount Unpaid Correlation: 37.8 %
```

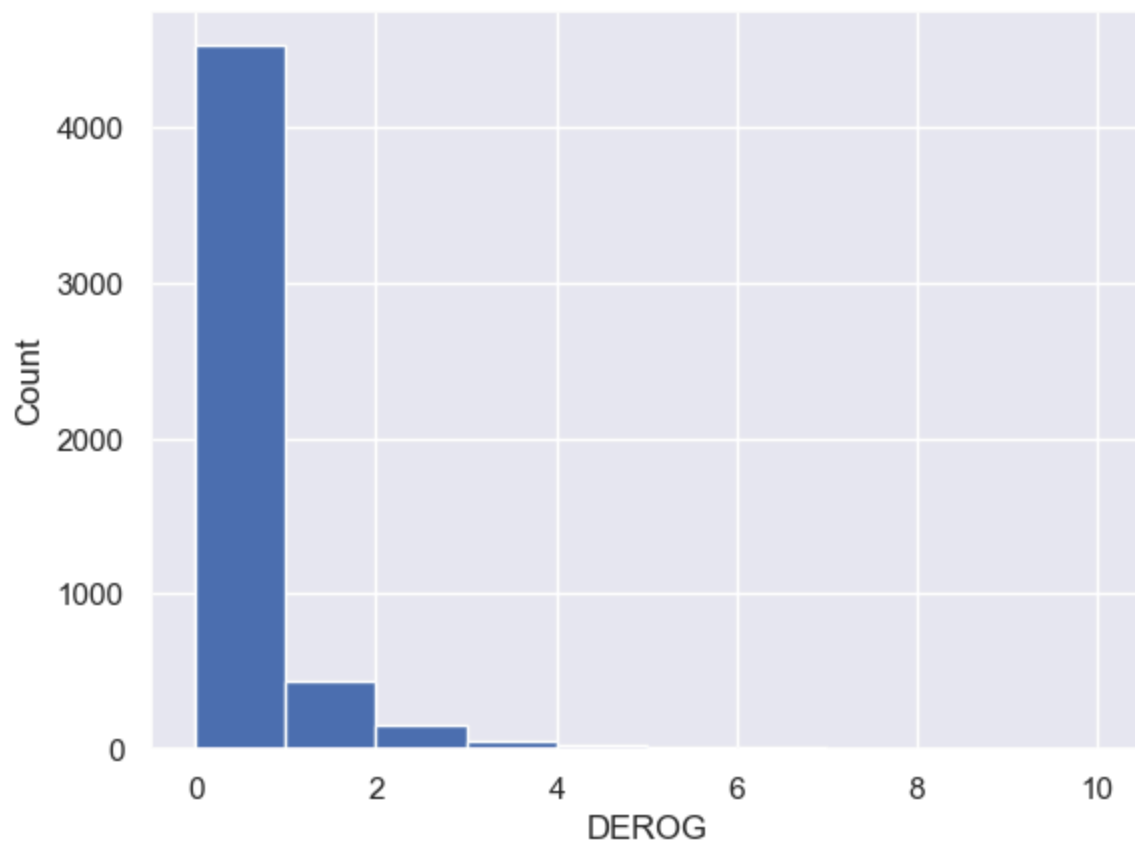
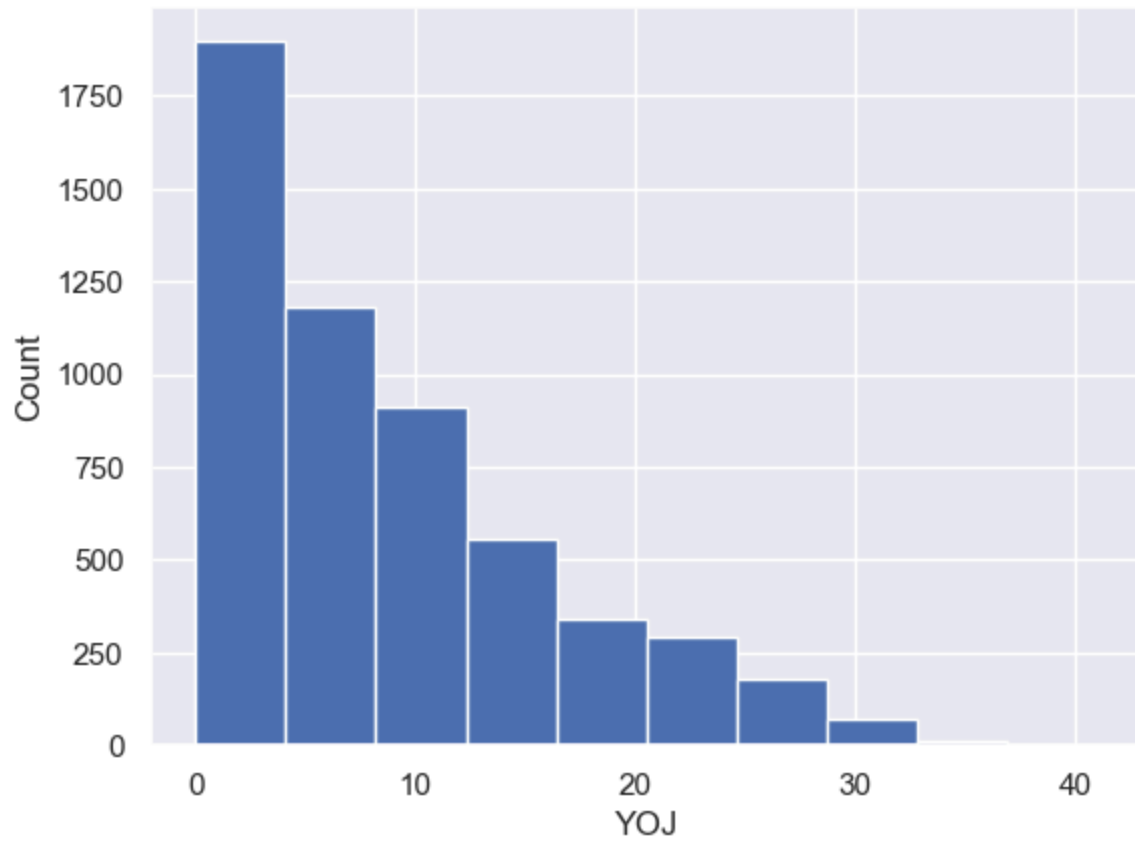
Exploring Distributions using Histograms

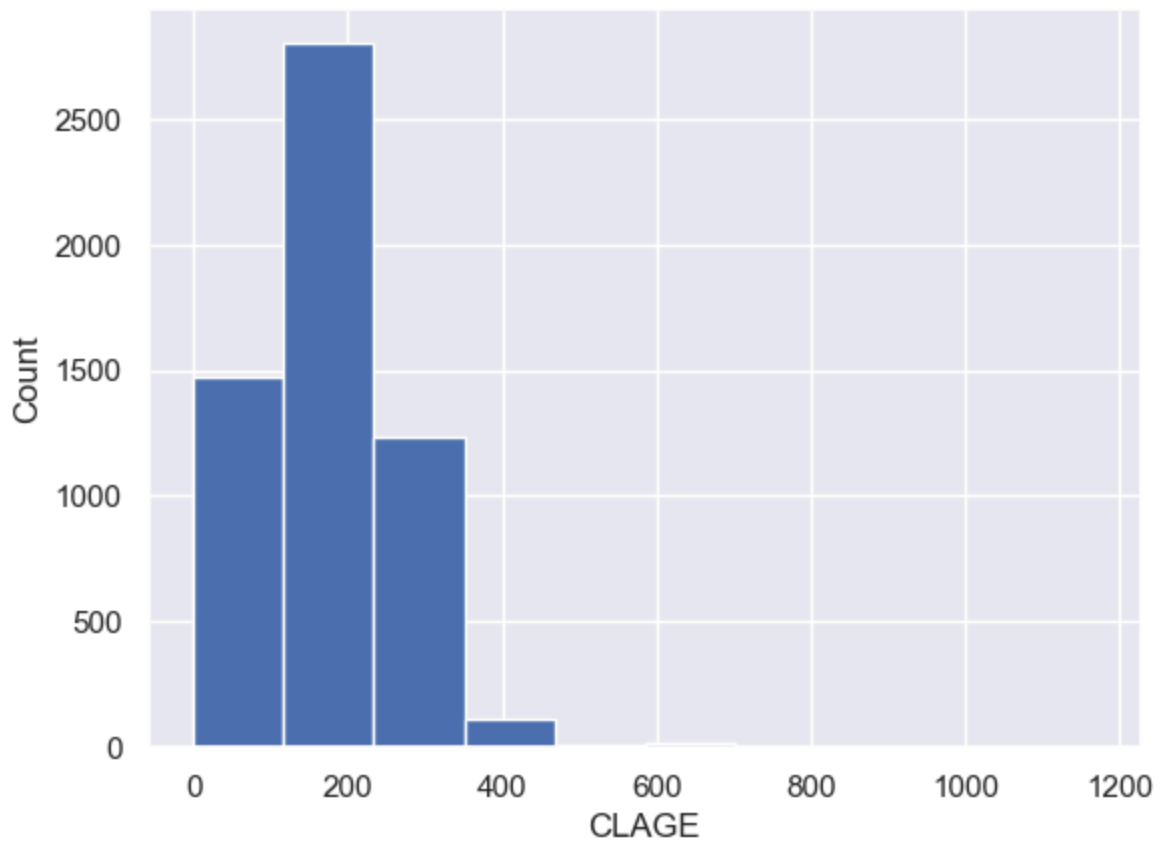
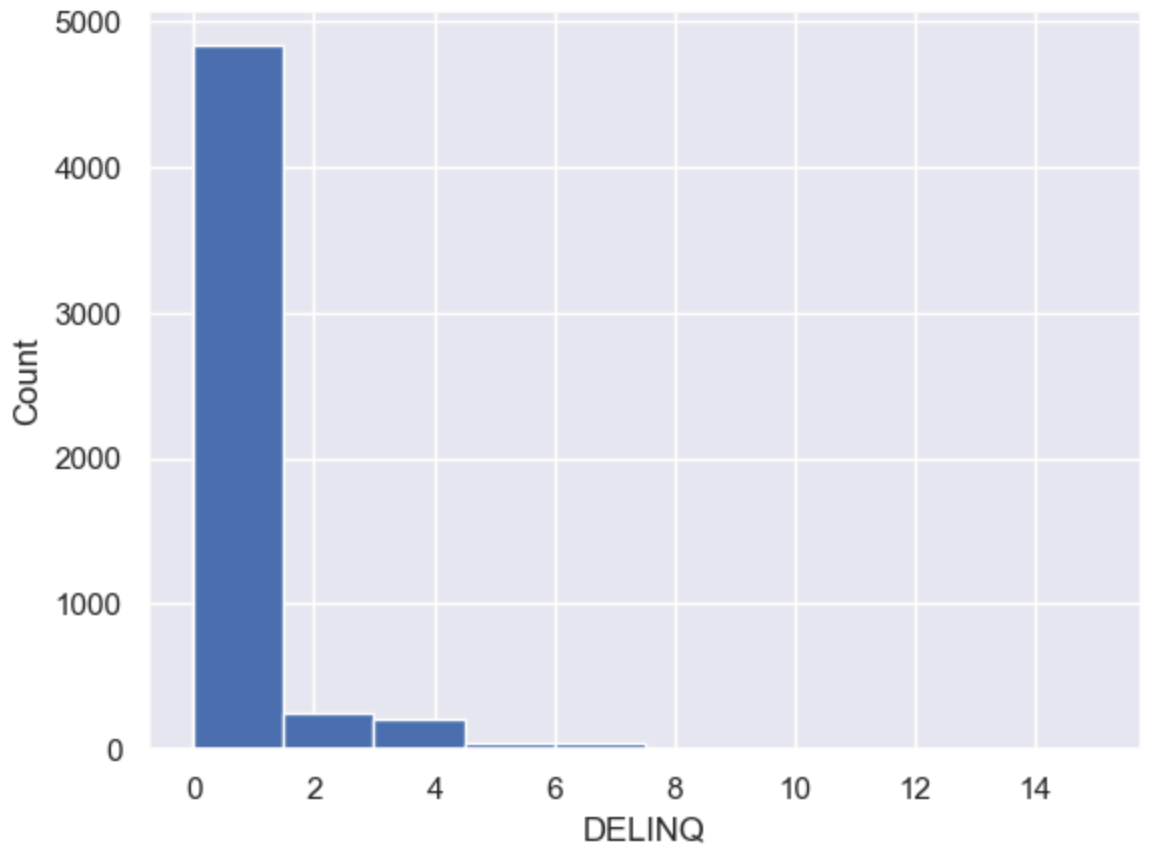
```
In [298... for i in intList:  
    plt.hist(df[i])  
    plt.xlabel(i)  
    plt.ylabel("Count")  
    plt.show()
```

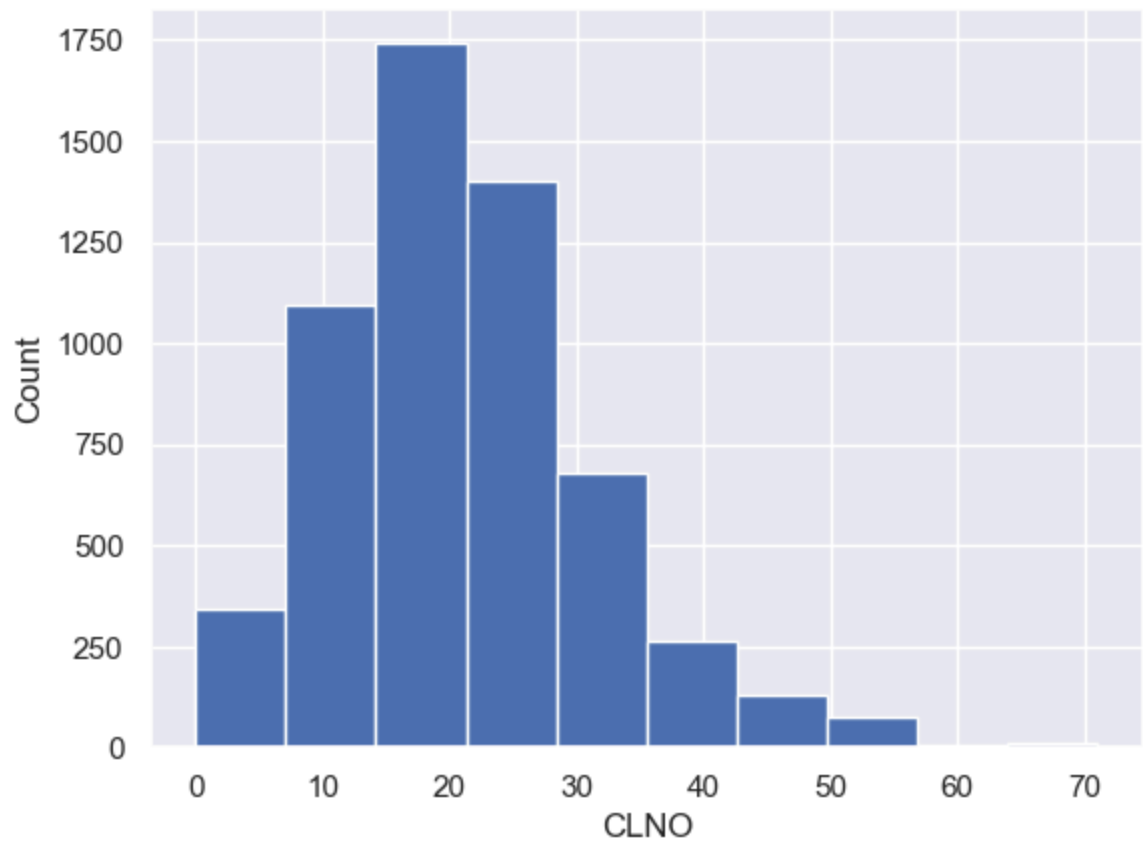
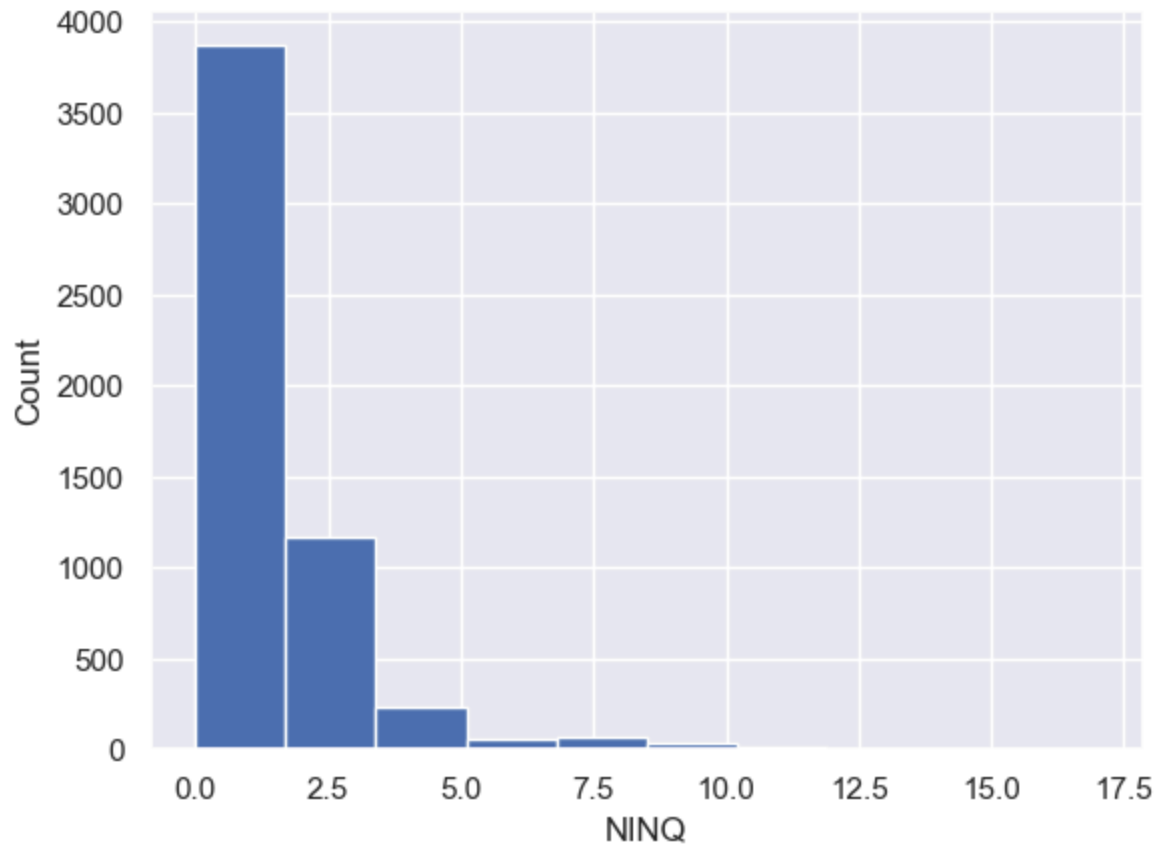


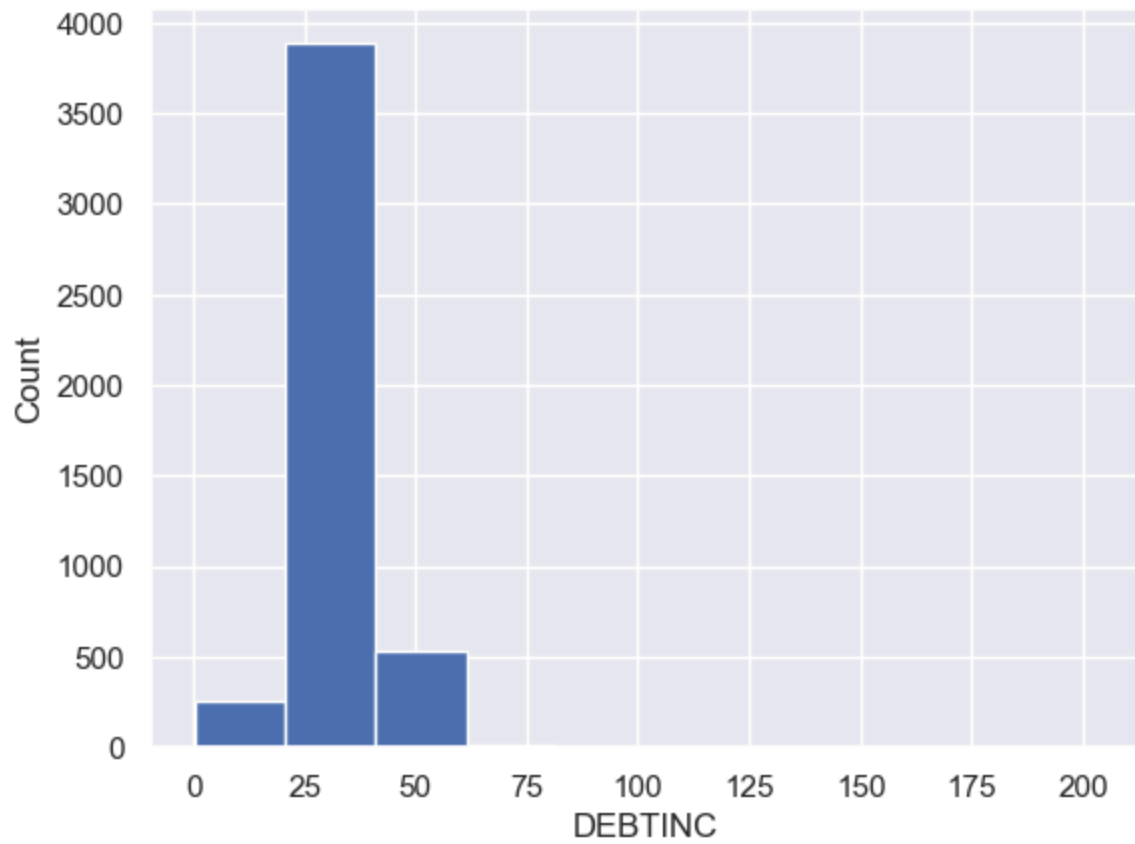
```
In [299... for i in floatList:  
    plt.hist(df[i])  
    plt.xlabel(i)  
    plt.ylabel("Count")  
    plt.show()
```





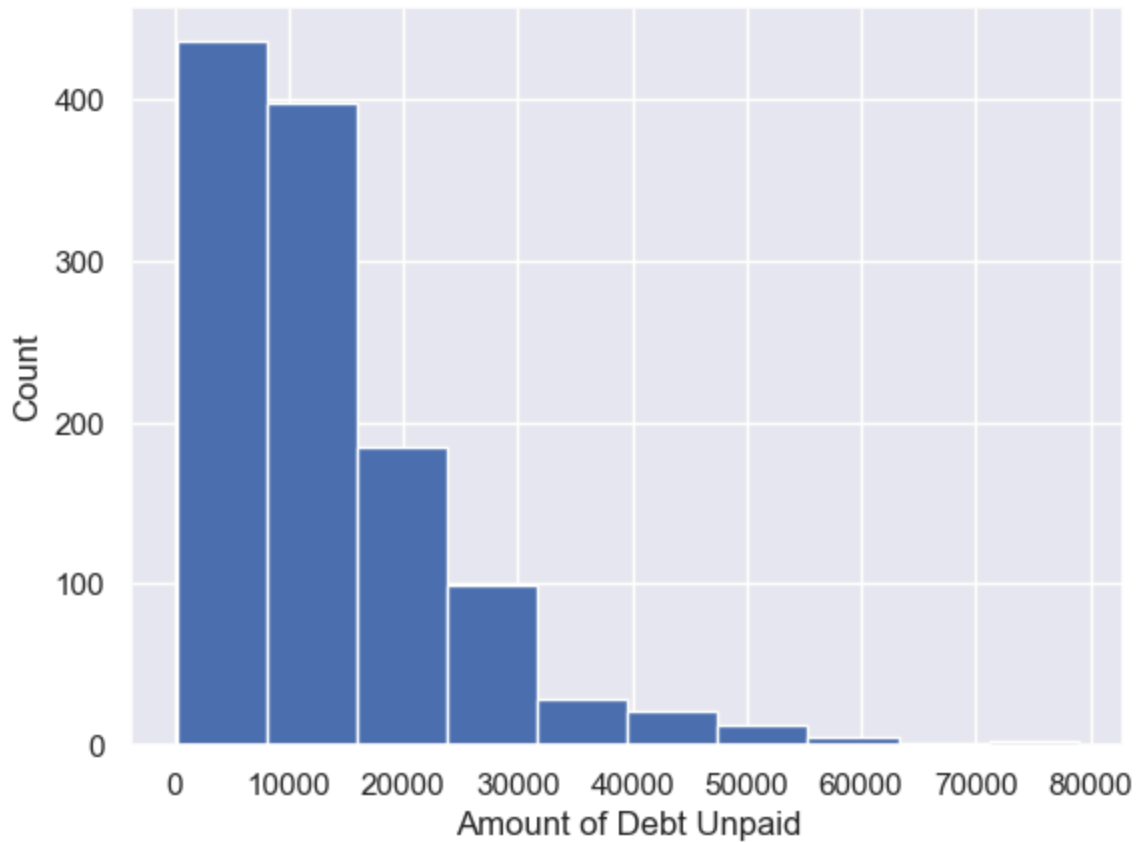






In [300...

```
plt.hist(df[TARGET_A])  
plt.xlabel("Amount of Debt Unpaid")  
plt.ylabel("Count")  
plt.show()
```



Exploring and Fixing Missing Values

```
In [301... #explore missing values in categorical variables
for i in objList:
    print(i)
    print(df[i].unique())
    g = df.groupby(i)
    print(g[i].count())
    print("MOST COMMON: ", df[i].mode()[0])
    print("MISSING: ", df[i].isna().sum())
    print("\n")
```

```

REASON
['HomeImp' nan 'DebtCon']
REASON
DebtCon    3928
HomeImp    1780
Name: REASON, dtype: int64
MOST COMMON: DebtCon
MISSING: 252

```

```

JOB
['Other' nan 'Office' 'Sales' 'Mgr' 'ProfExe' 'Self']
JOB
Mgr         767
Office      948
Other      2388
ProfExe    1276
Sales       109
Self        193
Name: JOB, dtype: int64
MOST COMMON: Other
MISSING: 279

```

In [302...

```

#replace missing values in categorical variables and make imputed variables
for i in objList:
    if df[i].isna().sum() == 0:continue
    print(i)
    print("HAS MISSING")
    Name = "IMP_" + i
    print(Name)
    df[Name] = df[i] #gives the copied variable the original variable's data
    df[Name] = df[Name].fillna("Missing")
    print("variable ",i," has this many missing: ", df[i].isna().sum())
    print("variable ",Name," has this many missing: ", df[i].isna().sum())
    g = df.groupby(Name)
    print(g[Name].count())
    print("\n")

```

```

REASON
HAS MISSING
IMP_REASON
variable REASON has this many missing: 252
variable IMP_REASON has this many missing: 252
IMP_REASON
DebtCon    3928
HomeImp    1780
Missing     252
Name: IMP_REASON, dtype: int64

```

```

JOB
HAS MISSING
IMP_JOB
variable JOB has this many missing: 279
variable IMP_JOB has this many missing: 279
IMP_JOB
Mgr         767
Missing     279
Office      948
Other       2388
ProfExe     1276
Sales       109
Self        193
Name: IMP_JOB, dtype: int64

```

```

In [303... #checking if missing values are successfully replaced
for i in ["JOB", "IMP_JOB"]:
    print(i)
    print(df[i].unique())
    g = df.groupby(i)
    print(g[i].count())
    print("MOST COMMON = ", df[i].mode()[0])
    print("MISSING = ", df[i].isna().sum())
    print("\n")

```

```

JOB
['Other' nan 'Office' 'Sales' 'Mgr' 'ProfExe' 'Self']
JOB
Mgr          767
Office       948
Other        2388
ProfExe      1276
Sales        109
Self         193
Name: JOB, dtype: int64
MOST COMMON = Other
MISSING = 279

```

```

IMP_JOB
['Other' 'Missing' 'Office' 'Sales' 'Mgr' 'ProfExe' 'Self']
IMP_JOB
Mgr          767
Missing      279
Office       948
Other        2388
ProfExe      1276
Sales        109
Self         193
Name: IMP_JOB, dtype: int64
MOST COMMON = Other
MISSING = 0

```

In [304...

#Pie Charts

```

for i in ["IMP_REASON", "IMP_JOB"]:
    x = df[i].value_counts(dropna = False)
    print(x)
    labels = x.axes[0].tolist()
    print(labels)
    slices = list(x)
    print(slices)
    plt.pie(slices, labels = labels, startangle = 90, autopct = "%1.1f%%")
    plt.title("Pie Chart of Object Variable: " + i)
    plt.show()

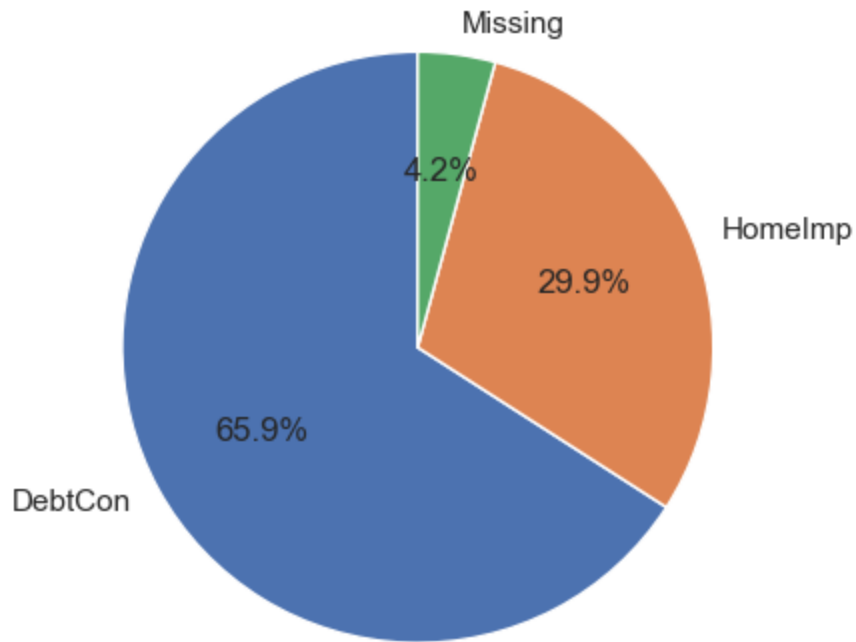
```

```

IMP_REASON
DebtCon      3928
HomeImp      1780
Missing       252
Name: count, dtype: int64
['DebtCon', 'HomeImp', 'Missing']
[3928, 1780, 252]

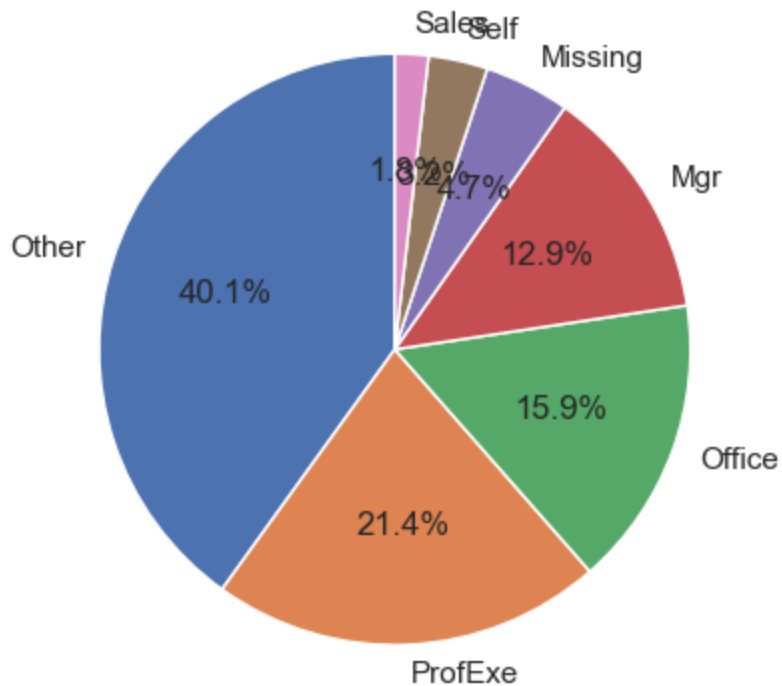
```


Pie Chart of Object Variable: IMP_REASON



```
IMP_JOB
Other      2388
ProfExe    1276
Office     948
Mgr        767
Missing    279
Self       193
Sales      109
Name: count, dtype: int64
['Other', 'ProfExe', 'Office', 'Mgr', 'Missing', 'Self', 'Sales']
[2388, 1276, 948, 767, 279, 193, 109]
```

Pie Chart of Object Variable: IMP_JOB



In [305... *#exploring and flagging missing values in numerical variables*

```
for i in intList:
    if df[i].isna().sum() == 0: continue
    Flag = "M_" + i
    Imp = "IMP_" + i
    df[Flag] = df[i].isna() + 0
    df[Imp] = df[i]
    df.loc[df[Imp].isna(), Imp] = df[i].median()
```

In [306...

```
for i in floatList:
    if df[i].isna().sum() == 0: continue
    Flag = "M_" + i
    Imp = "IMP_" + i
    df[Flag] = df[i].isna() + 0
    df[Imp] = df[i]
    df.loc[df[Imp].isna(), Imp] = df[i].median()
```

One Hot Encoding of Categorical Variables

In [307...

```
for i in ["IMP_REASON", "IMP_JOB"]:
    prefix = "z_" + i
    y = pd.get_dummies(df[i], prefix=prefix, dummy_na = 0, dtype=int)
    print(y.head().T)
    df = pd.concat([df, y], axis = 1)
    df = df.loc[:, ~df.columns.duplicated()]
```

	0	1	2	3	4
z_IMP_REASON_DebtCon	0	0	0	0	0
z_IMP_REASON_HomeImp	1	1	1	0	1
z_IMP_REASON_Missing	0	0	0	1	0
	0	1	2	3	4
z_IMP_JOB_Mgr	0	0	0	0	0
z_IMP_JOB_Missing	0	0	0	1	0
z_IMP_JOB_Office	0	0	0	0	1
z_IMP_JOB_Other	1	1	1	0	0
z_IMP_JOB_ProfExe	0	0	0	0	0
z_IMP_JOB_Sales	0	0	0	0	0
z_IMP_JOB_Self	0	0	0	0	0

Clean Up Final DataFrame

```
In [308... df_clean = df[sorted(df.columns)]

column = df_clean.pop(TARGET_F)
df_clean.insert(0, TARGET_F, column)

column = df_clean.pop(TARGET_A)
df_clean.insert(1, TARGET_A, column)

df_clean = df_clean.drop(columns = ["JOB", "REASON", "CLAGE", "CLNO", "DEBTINC", "DELI",
                                   "DEROG", "MORTDUE", "NINQ", "VALUE", "YOJ"], axis = 1)

print(df_clean.head().T)
```

	0	1	2	3	4
TARGET_BAD_FLAG	1	1	1	1	0
TARGET_LOSS_AMT	641.0	1109.0	767.0	1425.0	NaN
IMP_CLAGE	94.366667	121.833333	149.466667	173.466667	93.333333
IMP_CLNO	9.0	14.0	10.0	20.0	14.0
IMP_DEBTINC	34.818262	34.818262	34.818262	34.818262	34.818262
IMP_DELINQ	0.0	2.0	0.0	0.0	0.0
IMP_DEROG	0.0	0.0	0.0	0.0	0.0
IMP_JOB	Other	Other	Other	Missing	Office
IMP_MORTDUE	25860.0	70053.0	13500.0	65019.0	97800.0
IMP_NINQ	1.0	0.0	1.0	1.0	0.0
IMP_REASON	HomeImp	HomeImp	HomeImp	Missing	HomeImp
IMP_VALUE	39025.0	68400.0	16700.0	89235.5	112000.0
IMP_YOJ	10.5	7.0	4.0	7.0	3.0
LOAN	1100	1300	1500	1500	1700
M_CLAGE	0	0	0	1	0
M_CLNO	0	0	0	1	0
M_DEBTINC	1	1	1	1	1
M_DELINQ	0	0	0	1	0
M_DEROG	0	0	0	1	0
M_MORTDUE	0	0	0	1	0
M_NINQ	0	0	0	1	0
M_VALUE	0	0	0	1	0
M_YOJ	0	0	0	1	0
z_IMP_JOB_Mgr	0	0	0	0	0
z_IMP_JOB_Missing	0	0	0	1	0
z_IMP_JOB_Office	0	0	0	0	1
z_IMP_JOB_Other	1	1	1	0	0
z_IMP_JOB_ProfExe	0	0	0	0	0
z_IMP_JOB_Sales	0	0	0	0	0
z_IMP_JOB_Self	0	0	0	0	0
z_IMP_REASON_DebtCon	0	0	0	0	0
z_IMP_REASON_HomeImp	1	1	1	0	1
z_IMP_REASON_Missing	0	0	0	1	0

In [310... `print(df_clean.describe().T)`

	count	mean	std	min \
TARGET_BAD_FLAG	5960.0	0.199497	0.399656	0.000000
TARGET_LOSS_AMT	1189.0	13414.576955	10839.455965	224.000000
IMP_CLAGE	5960.0	179.440725	83.574697	0.000000
IMP_CLNO	5960.0	21.247819	9.951308	0.000000
IMP_DEBTINC	5960.0	34.000651	7.644528	0.524499
IMP_DELINQ	5960.0	0.405705	1.079256	0.000000
IMP_DEROG	5960.0	0.224329	0.798458	0.000000
IMP_MORTDUE	5960.0	73001.041812	42552.726779	2063.000000
IMP_NINQ	5960.0	1.170134	1.653866	0.000000
IMP_VALUE	5960.0	101540.387423	56869.436682	8000.000000
IMP_YOJ	5960.0	8.756166	7.259424	0.000000
LOAN	5960.0	18607.969799	11207.480417	1100.000000
M_CLAGE	5960.0	0.051678	0.221394	0.000000
M_CLNO	5960.0	0.037248	0.189386	0.000000
M_DEBTINC	5960.0	0.212584	0.409170	0.000000
M_DELINQ	5960.0	0.097315	0.296412	0.000000
M_DEROG	5960.0	0.118792	0.323571	0.000000
M_MORTDUE	5960.0	0.086913	0.281731	0.000000
M_NINQ	5960.0	0.085570	0.279752	0.000000
M_VALUE	5960.0	0.018792	0.135801	0.000000
M_YOJ	5960.0	0.086409	0.280991	0.000000
z_IMP_JOB_Mgr	5960.0	0.128691	0.334886	0.000000
z_IMP_JOB_Missing	5960.0	0.046812	0.211254	0.000000
z_IMP_JOB_Office	5960.0	0.159060	0.365763	0.000000
z_IMP_JOB_Other	5960.0	0.400671	0.490076	0.000000
z_IMP_JOB_ProfExe	5960.0	0.214094	0.410227	0.000000
z_IMP_JOB_Sales	5960.0	0.018289	0.134004	0.000000
z_IMP_JOB_Self	5960.0	0.032383	0.177029	0.000000
z_IMP_REASON_DebtCon	5960.0	0.659060	0.474065	0.000000
z_IMP_REASON_HomeImp	5960.0	0.298658	0.457708	0.000000
z_IMP_REASON_Missing	5960.0	0.042282	0.201248	0.000000

	25%	50%	75%	max
TARGET_BAD_FLAG	0.000000	0.000000	0.000000	1.000000
TARGET_LOSS_AMT	5639.000000	11003.000000	17634.000000	78987.000000
IMP_CLAGE	117.371430	173.466667	227.143058	1168.233561
IMP_CLNO	15.000000	20.000000	26.000000	71.000000
IMP_DEBTINC	30.763159	34.818262	37.949892	203.312149
IMP_DELINQ	0.000000	0.000000	0.000000	15.000000
IMP_DEROG	0.000000	0.000000	0.000000	10.000000
IMP_MORTDUE	48139.000000	65019.000000	88200.250000	399550.000000
IMP_NINQ	0.000000	1.000000	2.000000	17.000000
IMP_VALUE	66489.500000	89235.500000	119004.750000	855909.000000
IMP_YOJ	3.000000	7.000000	12.000000	41.000000
LOAN	11100.000000	16300.000000	23300.000000	89900.000000
M_CLAGE	0.000000	0.000000	0.000000	1.000000
M_CLNO	0.000000	0.000000	0.000000	1.000000
M_DEBTINC	0.000000	0.000000	0.000000	1.000000
M_DELINQ	0.000000	0.000000	0.000000	1.000000
M_DEROG	0.000000	0.000000	0.000000	1.000000
M_MORTDUE	0.000000	0.000000	0.000000	1.000000
M_NINQ	0.000000	0.000000	0.000000	1.000000
M_VALUE	0.000000	0.000000	0.000000	1.000000
M_YOJ	0.000000	0.000000	0.000000	1.000000
z_IMP_JOB_Mgr	0.000000	0.000000	0.000000	1.000000

z_IMP_JOB_Missing	0.000000	0.000000	0.000000	1.000000
z_IMP_JOB_Office	0.000000	0.000000	0.000000	1.000000
z_IMP_JOB_Other	0.000000	0.000000	1.000000	1.000000
z_IMP_JOB_ProfExe	0.000000	0.000000	0.000000	1.000000
z_IMP_JOB_Sales	0.000000	0.000000	0.000000	1.000000
z_IMP_JOB_Self	0.000000	0.000000	0.000000	1.000000
z_IMP_REASON_DebtCon	0.000000	1.000000	1.000000	1.000000
z_IMP_REASON_HomeImp	0.000000	0.000000	1.000000	1.000000
z_IMP_REASON_Missing	0.000000	0.000000	0.000000	1.000000

In []: